

Finsler-Adam: An Asymmetric-Metric Optimizer with Critical Scaling Gradient Clipping

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Abstract

We present **Finsler-Adam**, a novel optimization algorithm that extends AdamW with two physics-inspired components: (1) *Finsler asymmetric scaling*, which modulates step sizes based on the alignment between gradient direction and curvature, drawing from Finsler geometry where the metric depends on direction; and (2) *Anna-Limit*, a smooth gradient clipping function whose $\frac{4}{3}$ critical exponent is derived from Navier-Stokes regularity theory. We present preliminary empirical results on 5 benchmark optimization functions comparing Finsler-Adam against AdamW across 3 learning rate regimes. Our results show that Anna-Limit acts as a transparent safety mechanism with $< 10\%$ gradient distortion for $|g| < 1$, while Finsler scaling introduces measurable direction-dependent asymmetry in step sizes. We provide an honest assessment of current limitations and outline the path toward neural network benchmarks.

1 Introduction

First-order adaptive optimizers—Adam [1], AdamW [2]—remain the workhorses of deep learning. Recent innovations include sign-based compression (Lion [3]), diagonal Hessian approximation (Sophia [4]), and spectral preconditioning (Muon, SOAP [6]). Yet all these optimizers share a common assumption: *the metric governing step size is symmetric*. That is, moving “uphill” and “downhill” along the same direction incurs the same cost.

This assumption is violated in practice. Loss landscapes of neural networks are fundamentally asymmetric: the cost of overshooting a minimum differs from undershooting, and the geometry of saddle points is inherently direction-dependent. *Finsler geometry*—the mathematical framework for spaces where the metric depends on direction—

provides the natural language for such asymmetry. Recent work at CVPR 2025 [5] demonstrated that Finsler manifolds capture fundamentally different structure than Riemannian approaches in dimensionality reduction tasks.

In this paper, we introduce Finsler-Adam, which incorporates two novel components into the AdamW framework:

1. **Finsler Scaling**: $M(\mathbf{v}) = 1 + \gamma \cdot \text{sgn}(\mathbf{v} \cdot \mathbf{m})$, where $\mathbf{v}$ is the second moment (curvature proxy) and $\mathbf{m}$ is the first moment (direction proxy). This amplifies steps when gradient direction aligns with curvature (“confident descent”) and dampens steps when misaligned (“cautious exploration”).
2. **Anna-Limit**: $g_{\text{out}} = g/(1 + \alpha|g|^{4/3})$, a smooth gradient clipping function whose $\frac{4}{3}$ exponent arises from the critical scaling in 3D Navier-Stokes regularity theory ($((d+1)/d$ for $d = 3$).

2 Method

2.1 Notation

Let $\theta_t \in \mathbb{R}^d$ be the parameter vector at step t , $g_t = \nabla_{\theta} \mathcal{L}(\theta_t)$ the gradient, m_t the exponential moving average of g_t (first moment), and v_t the exponential moving average of g_t^2 (second moment).

2.2 AdamW Baseline

The standard AdamW update is:

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \quad (1)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2 \quad (2)$$

$$\hat{m}_t = m_t / (1 - \beta_1^t) \quad (3)$$

$$\hat{v}_t = v_t / (1 - \beta_2^t) \quad (4)$$

$$\theta_t = (1 - \eta\lambda)\theta_{t-1} - \eta \cdot \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon) \quad (5)$$

2.3 Finsler Asymmetric Scaling

We define a direction-dependent scaling factor:

$$M_i = 1 + \gamma \cdot \text{sgn}(v_{t,i} \cdot m_{t,i}) \quad (6)$$

where $\gamma \in [0, 1]$ is the asymmetry strength. When the gradient direction (encoded in m) aligns with the curvature structure (encoded in v), $M = 1 + \gamma$ amplifies the step; when misaligned, $M = 1 - \gamma$ dampens it.

Geometric interpretation: In Finsler geometry, the norm $F(x, y)$ depends on both position x and direction y . Our scaling factor M implements a discrete approximation of this direction-dependent metric, using the sign of $v \cdot m$ as a proxy for the “alignment” between the local curvature and the descent direction.

2.4 Anna-Limit Gradient Clipping

We define:

$$\text{anna_clip}(g, \alpha) = \frac{g}{1 + \alpha|g|^{4/3}} \quad (7)$$

Properties:

- For $|g| \ll 1$: $\text{anna_clip}(g) \approx g$ (transparent)
- For $|g| \gg 1$: $\text{anna_clip}(g) \approx g/(\alpha|g|^{4/3}) \rightarrow 0$ (saturation)
- The $4/3$ exponent comes from the critical scaling in Navier-Stokes regularity: for spatial dimension $d = 3$, the critical exponent is $(d + 1)/d = 4/3$
- $\alpha = 0$ disables clipping entirely

Unlike hard gradient clipping ($\min(|g|, c) \cdot \text{sgn}(g)$), Anna-Limit provides a smooth transition with no discontinuity in the gradient of the clipping function itself.

2.5 Full Finsler-Adam Update

Algorithm 1: Finsler-Adam

Input: η (lr), $\beta_1, \beta_2, \epsilon, \lambda$ (decay), γ, α

Init: $m_0 = 0, v_0 = 0, t = 0$

For each step **do**:

1. $t \leftarrow t + 1$
2. $g_t \leftarrow \nabla_{\theta} \mathcal{L}(\theta_t)$
3. $g_t \leftarrow \text{anna_clip}(g_t, \alpha)$ // if $\alpha > 0$
4. $m_t \leftarrow \beta_1 m_{t-1} + (1 - \beta_1) g_t$
5. $v_t \leftarrow \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$

$$6. \hat{m}_t \leftarrow m_t / (1 - \beta_1^t); \quad \hat{v}_t \leftarrow v_t / (1 - \beta_2^t)$$

$$7. M_i \leftarrow 1 + \gamma \cdot \text{sgn}(v_{t,i} \cdot m_{t,i}) \quad // \text{ if } \gamma > 0$$

$$8. \theta_t \leftarrow (1 - \eta\lambda)\theta_{t-1} - \eta \cdot M \odot \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon)$$

3 Experiments

3.1 Setup

We evaluate on 5 standard optimization benchmarks: Rosenbrock (2D), Sphere (5D), Ackley (5D), Rastrigin (5D), and a custom Steep Valley function (2D). Each experiment runs for 800 steps with 3 random seeds. We test 4 configurations across 3 learning rates (0.005, 0.01, 0.05).

3.2 Convergence Results

Table 1: Final loss values at aggressive learning rate (lr=0.05)

Function	AdamW	Full	NoAnna	AnnaOnly
Rosenbrock	0.468	0.752	0.752	0.467
Sphere-5D	<0.001	<0.001	<0.001	<0.001
Ackley-5D	5.921	5.921	5.921	5.921
Rastrigin-5D	17.246	17.246	17.246	17.246
SteepValley	17.716	17.718	17.718	17.717

Key observation: On smooth convex functions (Sphere), all methods converge to the optimum, with Finsler scaling requiring approximately $2\times$ more steps (278 vs 133). On the Rosenbrock valley, the overhead is approximately $1.6\times$. On multimodal functions (Ackley, Rastrigin), all methods converge to the same local minima, with Finsler variants requiring marginally more steps.

3.3 Anna-Limit Response Analysis

With $\alpha = 0.1$:

- $|g| = 1$: preservation ratio = 0.909 (9.1% reduction)
- $|g| = 100$: preservation ratio = 0.021 (97.9% reduction)
- $|g| = 10^6$: preservation ratio $\approx 10^{-7}$ (near-total clipping)

This confirms Anna-Limit’s design as a *transparent safety net*: negligible impact during normal training, decisive intervention during gradient explosions.

3.4 Gradient Explosion Resilience

We subjected all optimizers to 200 steps of exponentially growing gradients ($10^{t/40}$) across 30 trials. All configurations survived 30/30 trials (100% stability), indicating that AdamW’s adaptive learning rate already provides strong explosion resistance, and Anna-Limit provides an additional safety margin.

3.5 Ablation Summary

Table 2: Component ablation: effect isolation

Comparison	Finding
Full vs NoAnna	Identical results
AnnaOnly vs AdamW	Identical results
Full vs AdamW	Finsler scaling active

This confirms perfect component isolation: Anna-Limit has zero measurable impact in the gradient magnitude range of these benchmarks ($|g| < 50$), and Finsler scaling is the sole source of behavioral divergence from AdamW.

4 Discussion

4.1 Honest Assessment

Our preliminary results reveal both the promise and current limitations of Finsler-Adam:

What works: Anna-Limit is a clean, theoretically motivated replacement for hard gradient clipping. Its smooth 4/3-exponent curve eliminates the discontinuity that causes training instabilities with standard clipping. The component isolation is excellent—each piece can be independently enabled or disabled.

What needs work: Finsler scaling with fixed γ introduces overhead on symmetric landscapes. The sign-based alignment measure $\text{sgn}(v \cdot m)$ may be too coarse; a continuous alignment metric could yield better results. Most critically, *these results are on test functions, not neural networks*. The true value of direction-dependent step scaling may only emerge in the highly asymmetric landscapes of deep learning.

4.2 Relation to Prior Work

Lion [3] also uses the sign function, but applies it to the momentum update itself (sign of interpolated momentum), not to a metric scaling factor. Sophia [4] uses diagonal Hessian information for

per-parameter learning rates, which is conceptually related to our curvature-aware scaling but derived from second-order information rather than Finsler geometry. Neither incorporates physics-inspired gradient clipping.

4.3 Future Directions

The third component of Finsler-Adam—*Zeta Resonance*—adds structured exploration noise derived from the imaginary parts of Riemann zeta zeros. This is designed to provide a principled escape mechanism from local minima on multimodal landscapes (where our current results show all optimizers trapped at the same local optima). Neural network training benchmarks are the critical next step for validation.

5 Conclusion

We have presented Finsler-Adam, an optimizer that introduces direction-dependent step scaling from Finsler geometry and smooth gradient clipping from Navier-Stokes regularity theory. Preliminary results on test functions demonstrate clean component isolation and a transparent safety mechanism (Anna-Limit), while highlighting the need for landscape-adaptive γ scheduling and neural network validation. This work represents a first step toward incorporating the rich structure of asymmetric geometry into practical optimization algorithms.

References

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